

Embeddings: meaning as geometry

1 Words need numbers

IDs and one-hot slots are numbers, but they carry no meaning. “invoice” is as far from “bill” as from “trojan”.

2 Know a word by its company

Words that appear in the same contexts get similar vectors. Nobody labels them: counting does it.

3 Meaning becomes geometry

Similar meaning = small angle. Cosine similarity measures it, from about -1 to 1.



4 Alerts can be embedded too

Whole alerts become vectors. Similar alerts land together, which powers search, de-duplication and RAG.

5 Neighbours give probabilities

“Among the 50 most similar past alerts, how many were attacks?” is honest (ECE 0.015), but blunt (AUC 0.68).

6 Meaning isn’t the numbers

Text similarity knows what an alert is about, not its threat score. Structured fields scored AUC 0.89. Keep the numbers as numbers.